

Topic: Instruction Formats

Q. What is an Instruction Format? Explain the different types of instruction formats based on the number of addresses (Zero, One, Two, and Three-Address instructions) with examples.

****Definition to Remember****

An ****Inst**
perform
have ze

Instruction Formats & Addressing

Used in Stack-based architectures. The operands are implicitly located at the top of the stack.

* ****For**

2. One-Address Instruction

Used in Accumulator-based architectures. One operand is specified; the second operand and the destination are implicitly the Accumulator register.

* ****For**

3. Two-Address Instruction

Used in General Purpose Register architectures (like Intel x86). It specifies two addresses, where one usually acts as both a source and the destination.

* ****For**

4. Three-Address Instruction

Used in modern RISC architectures (like ARM, MIPS). It specifies three distinct locations (one destination, two sources). Makes expressions short but requires longer instruction strings.

* ****For**

****Must-Write Points (for 10 marks)****

1. ****Inst**

****Quick Recall****

`Instruc
ADD R1